

QUESTION NO#1:

```
#include <iostream>
```

```
using namespace std;
```

```
class Shape {
```

```
protected:
```

```
    int numberOfSides;
```

```
    double area;
```

```
public:
```

```
    Shape(int n, double a) {
```

```
        numberOfSides = n;
```

```
        area = a;
```

```
    }
```

```
    virtual void generateArea() = 0;
```

```
};
```

```
class Rectangle : public Shape {
```

```
protected:
```

```
    double length, width;
```

```
public:
```

```
    Rectangle(double l, double w) : Shape(4, 0.0) {
```

```
        length = l;
```

```
        width = w;
```

```
    }
```

```
    void generateArea() {
```

```
        area = length * width;
```

```
        cout << "Rectangle Area: " << area << endl;
    }
};
```

```
class Circle : public Shape {
    double radius;
public:
    Circle(double r) : Shape(0, 0.0) {
        radius = r;
    }
    void generateArea() {
        area = 3.14159 * radius * radius;
        cout << "Circle Area: " << area << endl;
    }
};
```

```
class Triangle : public Shape {
    double height, base;
public:
    Triangle(double h, double b) : Shape(3, 0.0) {
        height = h;
        base = b;
    }
    void generateArea() {
        area = 0.5 * base * height;
        cout << "Triangle Area: " << area << endl;
    }
};
```

```
    }  
};  
  
class Square : public Rectangle {  
public:  
    Square(double side) : Rectangle(side, side) {}  
    void checkSides() {  
        if (length == width) {  
            cout << "It is a valid square." << endl;  
        }  
    }  
    void generateArea() {  
        area = length * width;  
        cout << "Square Area: " << area << endl;  
    }  
};  
  
int main() {  
    Shape* s;  
  
    Rectangle rect(10, 5);  
    Circle circ(4);  
    Triangle tri(4, 6);  
    Square sq(5);  
  
    s = &rect;
```

```
s->generateArea();

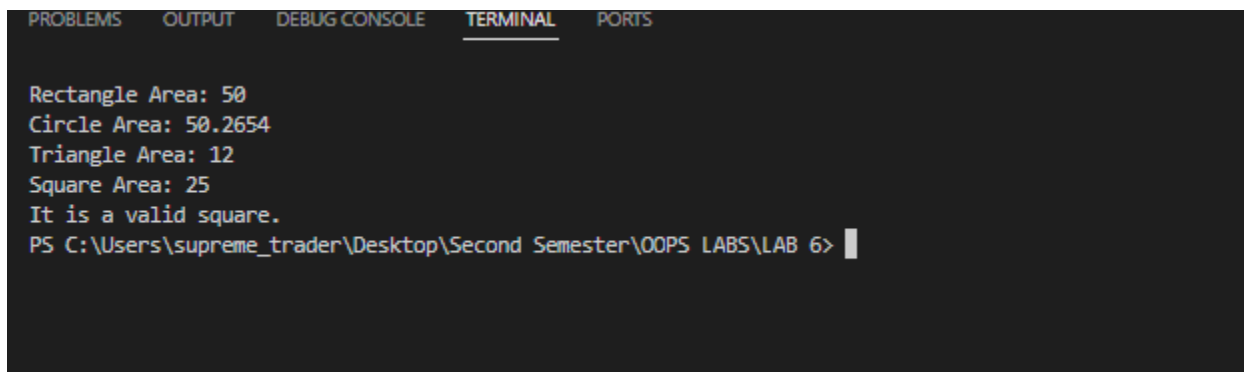
s = &circ;
s->generateArea();

s = &tri;
s->generateArea();

s = &sq;
s->generateArea();
sq.checkSides();

return 0;
}
```

OUTPUT:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Rectangle Area: 50
Circle Area: 50.2654
Triangle Area: 12
Square Area: 25
It is a valid square.
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 6>
```

QUESTION NO#2:

```
#include <iostream>
```

```
using namespace std;
```

```
class Calculator {
```

```
private:
```

```
    double Num1, Num2, Num3;
```

```
public:
```

```
    Calculator() {
```

```
        cout << "Enter number 1: ";
```

```
        cin >> Num1;
```

```
        cout << "Enter number 2: ";
```

```
        cin >> Num2;
```

```
        cout << "Enter number 3: ";
```

```
        cin >> Num3;
```

```
    }
```

```
    double getNum1() { return Num1; }
```

```
    double getNum2() { return Num2; }
```

```
    double getNum3() { return Num3; }
```

```
    double add(double a, double b) { return a + b; }
```

```
    double add(double a, double b, double c) { return a + b + c; }
```

```
    double multiply(double a, double b) { return a * b; }
```

```
    double multiply(double a, double b, double c) { return a * b * c; }
```

```
double subtract(double a, double b) { return a - b; }
```

```
double subtract(double a, double b, double c) { return a - b - c; }
```

```
double divide(double a, double b) {
```

```
    if (b == 0) {
```

```
        cout << "Error: Division by zero" << endl;
```

```
        return 0;
```

```
    }
```

```
    return a / b;
```

```
}
```

```
double divide(double a, double b, double c) {
```

```
    if (b == 0 || c == 0) {
```

```
        cout << "Error: Division by zero" << endl;
```

```
        return 0;
```

```
    }
```

```
    return a / b / c;
```

```
}
```

```
};
```

```
int main() {
```

```
    Calculator calc;
```

```
    double n1 = calc.getNum1();
```

```
    double n2 = calc.getNum2();
```

```
double n3 = calc.getNum3();

cout << calc.add(n1, n2) << endl;
cout << calc.add(n1, n2, n3) << endl;

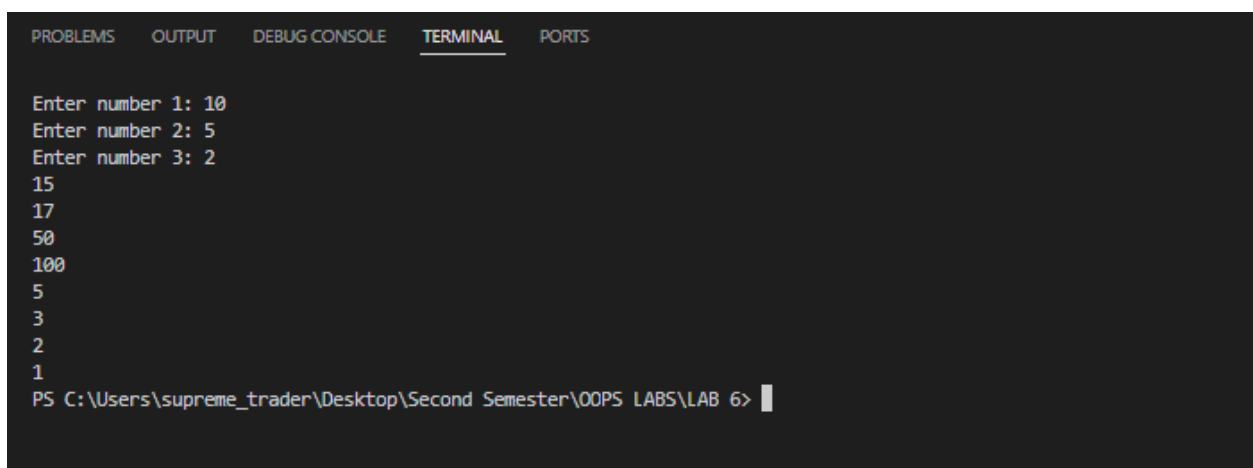
cout << calc.multiply(n1, n2) << endl;
cout << calc.multiply(n1, n2, n3) << endl;

cout << calc.subtract(n1, n2) << endl;
cout << calc.subtract(n1, n2, n3) << endl;

cout << calc.divide(n1, n2) << endl;
cout << calc.divide(n1, n2, n3) << endl;

return 0;
}
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Enter number 1: 10
Enter number 2: 5
Enter number 3: 2
15
17
50
100
5
3
2
1
15
17
50
100
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 6>
```

QUESTION NO#3:

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
class Vector {
```

```
private:
```

```
    double x, y;
```

```
public:
```

```
    Vector(double x = 0.0, double y = 0.0) {
```

```
        this->x = x;
```

```
        this->y = y;
```

```
    }
```

```
    Vector operator+(const Vector& obj) const {
```

```
        return Vector(x + obj.x, y + obj.y);
```

```
    }
```

```
    Vector operator-(const Vector& obj) const {
```

```
        return Vector(x - obj.x, y - obj.y);
```

```
    }
```

```
    Vector operator*(double scalar) const {
```

```
        return Vector(x * scalar, y * scalar);
```



```
}
```

```
Vector operator/(double scalar) const {
```

```
    if (scalar == 0) {
```

```
        cout << "Error: Division by zero" << endl;
```

```
        return Vector(0, 0);
```

```
    }
```

```
    return Vector(x / scalar, y / scalar);
```

```
}
```

```
double magnitude() const {
```

```
    return sqrt(x * x + y * y);
```

```
}
```

```
void display() const {
```

```
    cout << "(" << x << ", " << y << ")" << endl;
```

```
}
```

```
};
```

```
int main() {
```

```
    Vector v1(10.5, 5.5);
```

```
    Vector v2(2.5, 1.5);
```

```
    (v1 + v2).display();
```

```
    (v1 - v2).display();
```

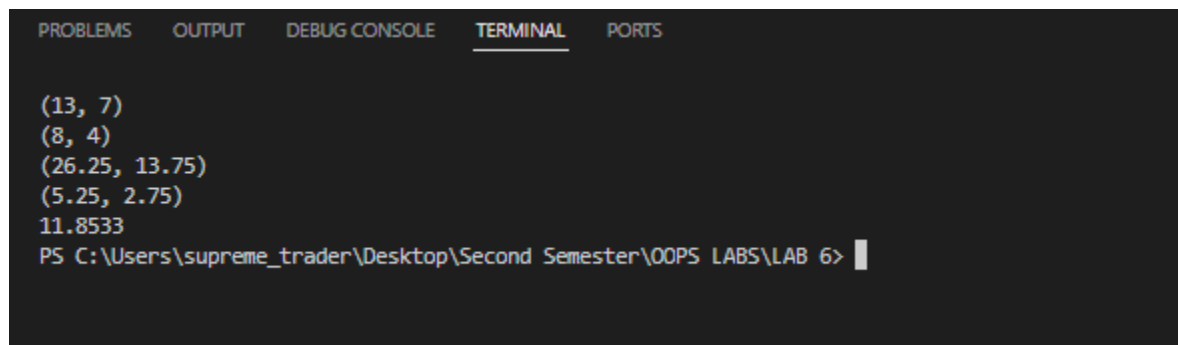
```
    (v1 * 2.5).display();
```

```
(v1 / 2.0).display();

cout << v1.magnitude() << endl;

return 0;
}
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

(13, 7)
(8, 4)
(26.25, 13.75)
(5.25, 2.75)
11.8533
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 6> |
```

QUESTION NO#4:

```
#include <iostream>

#include <string>

using namespace std;

class Person {

protected:

    string name;
```

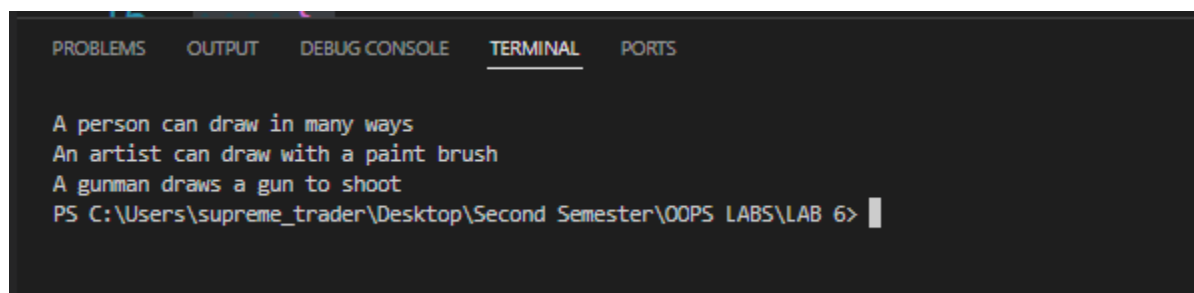
```
    string occupation;
public:
    Person(string _occupation) {
        occupation = _occupation;
    }
    virtual ~Person() {}
    virtual void Draw() {
        cout << "A person can draw in many ways" << endl;
    }
};

class Artist : public Person {
public:
    Artist() : Person("Artist") {}
    void Draw() {
        cout << "An artist can draw with a paint brush" << endl;
    }
};

class Gunman : public Person {
public:
    Gunman() : Person("Gunman") {}
    void Draw() {
        cout << "A gunman draws a gun to shoot" << endl;
    }
};
```

```
int main() {  
    Person* arr[3];  
  
    arr[0] = new Person("Unknown");  
    arr[1] = new Artist();  
    arr[2] = new Gunman();  
  
    for (int i = 0; i < 3; i++) {  
        arr[i]->Draw();  
    }  
  
    for (int i = 0; i < 3; i++) {  
        delete arr[i];  
    }  
  
    return 0;  
}
```

OUTPUT:



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there is a tab bar with five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. Below the tabs, the terminal displays the output of the program: 'A person can draw in many ways', 'An artist can draw with a paint brush', and 'A gunman draws a gun to shoot'. At the bottom of the terminal, the command prompt shows the file path 'PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 6>' followed by a cursor.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
A person can draw in many ways  
An artist can draw with a paint brush  
A gunman draws a gun to shoot  
PS C:\Users\supreme_trader\Desktop\Second Semester\OOPS LABS\LAB 6> |
```